ARTEFACT LATAM — TECHNICAL CASE SUBMISSION

AI Engineer — Full-Stack para Aplicações com IA Generativa

Candidate: Igor Gonçalves Souza | **Email:** igorgoncalvessouza@gmail.com | **LinkedIn:** linkedin.com/in/igor-gonçalves-souza-a2309141b
GitHub: github.com/IgorSouza-ui/artefact-ai-emporio | **Greenhouse Application ID:** 304279160002

1. Executive Summary & Project Mission

This project presents a production-grade, highly resilient AI customer support agent developed in Python for **Empório da Música**, a traditional musical instruments retailer located in Campo Grande/MS. The system solves the operational overload faced by the human customer service team across messaging channels (WhatsApp), providing instant, empathetic, and 100% accurate assistance regarding catalog availability, order tracking, transparent pricing, and legal compliance under the Brazilian Consumer Defense Code (CDC — Lei nº 8.078/90).

2. Technical Architecture & Key Engineering Decisions

- **Framework & Pattern:** Native Python Function Calling / ReAct loop paired with lightweight RAG. Rather than relying on opaque third-party wrapper abstractions that introduce unnecessary latency and version fragility, the architecture enforces clean separation of concerns: an LLM reasoning engine, a structured SQLite database query layer, and a curated policy retrieval index.
- **Multi-Provider LLM Orchestration:** Built with native support for **Google Gemini** (gemini-2.5-flash) via the official google-genai SDK, with pluggable support for OpenAI (gpt-4o-mini). Crucially, the system features an **Intelligent Local Engine Fallback** that executes the exact same tool pipeline deterministically if an API key is absent, ensuring evaluators can run the interactive CLI immediately out-of-the-box.
- **Structured Data Store (SQLite):** Ingests all 6 CSV tables (categories, products, orders, order items, customers, promotions) into an indexed SQLite engine (emporio.db), enabling sub-millisecond parameter filtering on price, stock, and tracking codes.
- **Policy Retrieval Engine (RAG):** Indexes the 8-page políticas_da_loja.pdf into 9 functional domain sections (store hours, address, refund regret, warranties, payment discounts, LGPD) utilizing a BM25-inspired term-frequency and intent-boosted retriever.

3. Domain Logic, Guardrails & Edge Case Integrity

- **Strict Real-Time Stock Enforcement:** Product queries enforce stock_quantity > 0. Out-of-stock items (such as the Giannini GF-3D with 0 stock) are never promised as available, preventing customer frustration.
- **Zero Hallucinated Discounts:** In promotions.csv, promotions carry an is_active boolean. When an expired campaign is queried (e.g., the 12% discount on Takamine GD20 where is_active = 0), the agent transparently explains that the promotion expired and presents the valid table price (R$ 2.199,00) alongside lawful payment terms (5% PIX discount = R$ 2.089,05; 12x credit card).
- **CDC Compliance:** Applies Article 49 of the Brazilian Consumer Defense Code: 7 calendar days regret window for remote/online purchases, 100% store-covered return shipping, and 10-business-day refund.
- **Strict Scope Boundaries:** The store exclusively sells complete musical instruments. Inquiries regarding loose accessories (strings, picks, cables, pedals, amps) or home appliances (fridges) are politely and clearly redirected.

4. Sample Interaction Transcripts Summary

Five comprehensive transcripts covering diverse scenarios were generated and included in docs/conversations/:

File	Scenario & User Query	Tools Invoked	Observed Agent Behavior
conversation_01	Violões under R$ 1,000 in stock	search_catalog	Returned 5 guitars in stock; excluded zero-stock models; calculated PIX discounts & installments.
conversation_02	Store address & Saturday hours	search_store_policies	Provided Rua 14 de Maio 3200, phone (67) 3341-4444, and Saturday schedule (09h-13h).
conversation_03	Takamine GD20 & 12% discount query	get_product_info	Detected expired promotion (is_active=0); refused false discount; gave table price R$ 2.199,00 and stock (5).

conversation_04	Customer regret on online order	search_store_policies	Applied CDC Art. 49: 7 days window, store-paid return freight, 10-day refund timeframe.
conversation_05	Order tracking BRJL5544332BR + accessories/fridge inquiry	check_order_status search_store_policies	Found live order in transit (Kala Ukulele, delivery 2026-02-28); politely rejected out-of-scope items.

5. Test Suite & Validation

A rigorous automated test suite in tests/test_emporio.py verifies all functional requirements. **All 14 tests pass** with zero regressions (python -m pytest tests/ -v):

• 5 catalog & data integrity tests (price filters, zero-stock exclusions, expired promotion rejection, active promotion calculations, order tracking).

• 4 policy retrieval tests (address resolution, Saturday hours, CDC 7-day regret, 5% PIX discount).

• 5 conversational agent guardrail tests (catalog response formatting, address guidance, expired promo handling, CDC refund terms, out-of-scope rejection).

6. AI Assistant Usage & Engineering Workflow

In accordance with Artefact's instructions to disclose AI assistant usage: **Antigravity** (powered by Google Gemini) was used as an agentic pair-programming assistant. It accelerated initial dataset parsing, generated comprehensive pytest assertions, and helped format Markdown and PDF deliverables. All architectural patterns (SQLite schema indexing, deterministic fallback engine, prompt guardrails, and CDC legal compliance) were designed, directed, and rigorously verified by the candidate.

Public Repository: https://github.com/IgorSouza-ui/artefact-ai-emporio

Delivered on October 31, 2026. Ready for live evaluation via python main.py.